

LeadFlow Autopilot

Automated Multi-Channel Lead Engagement Pipeline with AI Voice Calling

Form Intake • Email & WhatsApp Outreach • AI Voice Qualification • CRM Sync

Project Overview

LeadFlow Autopilot is a fully automated lead engagement pipeline that captures inbound form submissions and instantly activates a multi-channel outreach sequence spanning email, WhatsApp, and AI-powered voice calls. The system qualifies leads through an intelligent voice agent, extracts structured data from conversations, and syncs all results into Salesforce CRM in real time.

Built on n8n workflow automation with VAPI voice AI integration, the pipeline eliminates the delay between lead capture and first contact, ensuring every prospect receives personalized engagement within seconds of submitting their inquiry.

Attribute	Details
Role	AI Automation Architect
Industry	Free Zone / Business Licensing (UAE)
Platform	n8n (Self-Hosted Workflow Automation)
Voice AI	VAPI (Outbound AI Voice Calling)
Channels	Email (Gmail), WhatsApp Business API, Voice (VAPI)
CRM	Salesforce
Form Intake	Website Contact Form (Webhook Integration)
Storage	Google Drive (Call Recordings)

System Architecture

The pipeline follows a sequential, event-driven architecture. A single form submission triggers an orchestrated engagement sequence that spans three communication channels and includes automated call monitoring, retry logic, and structured data extraction.

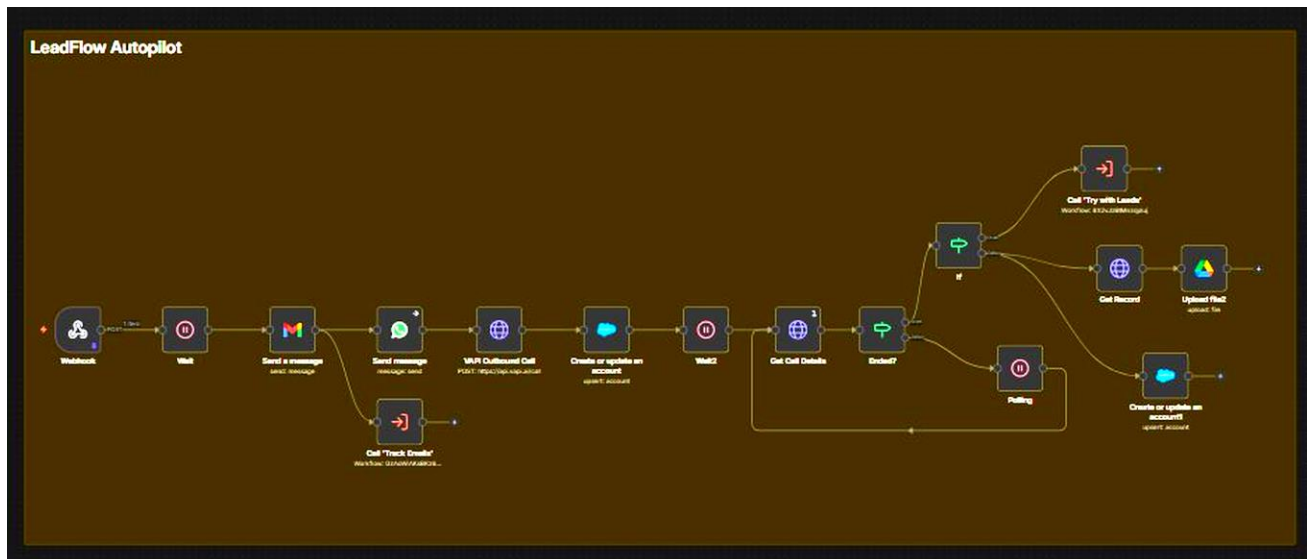


Figure 1: n8n Workflow — LeadFlow Autopilot Pipeline

End-to-End Flow

1. **Form Submission Received** — A webhook captures the lead's form data from the website contact form, including name, email, phone, enquiry type, and message.
2. **Timed Delay** — A configurable wait node introduces a brief delay before outreach begins, preventing the prospect from feeling overwhelmed by instant automated contact.
3. **Email Outreach** — Gmail sends a personalized welcome email acknowledging the prospect's interest and offering assistance.
4. **WhatsApp Message** — A WhatsApp Business API message is sent simultaneously, establishing a second touchpoint with the same personalized greeting.
5. **Email Tracking Workflow** — A sub-workflow is triggered to monitor email opens, clicks, and engagement metrics for the outreach email.
6. **AI Voice Call** — VAPI initiates an outbound AI voice call to the prospect's phone number, passing context variables (name, email, enquiry type) so the voice agent can personalize the conversation.
7. **CRM Lead Creation** — The lead record is created or updated in Salesforce with initial data and engagement status flags.
8. **Call Status Polling** — The workflow polls VAPI's API every 10 seconds to check if the call has ended, implementing a retry loop until completion is confirmed.
9. **Call Outcome Routing** — Once the call ends, the system evaluates the outcome: if the customer was busy or didn't answer, a follow-up retry workflow is triggered; if the call was answered, the recording and structured data are captured.
10. **Recording Archival** — The call recording is downloaded from VAPI and uploaded to Google Drive in a dedicated folder, named with the lead's details for easy retrieval.

11. **Structured Data Extraction** — VAPI’s structured output feature extracts key qualification data from the conversation (business activities, package interest, budget, visa needs, etc.) and the workflow updates Salesforce with these enriched fields.

Multi-Channel Engagement Strategy

The system implements a triple-channel engagement approach, ensuring maximum reach and response rates by contacting prospects through their preferred communication method.

Channel	Trigger	Content	Purpose
Email	Immediate after delay	Personalized welcome + offer to assist	Professional first touch; trackable
WhatsApp	Parallel with email	Friendly greeting + availability	High open-rate instant channel
Voice Call	After WhatsApp sent	AI-driven qualification conversation	Deep qualification + rapport

AI Voice Agent Integration

The VAPI voice agent is the core qualification engine of this pipeline. It conducts natural-sounding outbound calls with full context awareness, extracting structured business intelligence from each conversation.

Context Passing

The workflow passes five context variables to the voice agent before each call, enabling personalized conversation from the first second:

- First Name and Last Name — for natural greeting
- Email — for reference during the conversation
- Phone Number — the dialing target
- Enquiry Type — so the agent understands the prospect’s intent category

Structured Data Extraction

VAPI’s structured output blocks automatically extract the following fields from each completed call, which are then mapped directly to Salesforce fields:

Extracted Field	Description
Business Activities	What the prospect intends to do
Package Interest	Which licensing tier they prefer
License Period	Duration preference (1–10 years)

Office Space Requirements	Workspace needs
Visa Requirements	Number of visas needed
Budget	Stated or implied budget range
Company Name	Prospect's existing or planned company
Add-Ons Interest	Additional services requested
Business Category	B2B / B2C / B2G classification
Lead Status	AI-determined qualification level
Call Successful	Whether the call achieved its objective

Call Monitoring & Retry Logic

The system implements intelligent call lifecycle management through a polling mechanism and conditional retry logic.

Polling Mechanism

After initiating the VAPI call, the workflow enters a polling loop that checks the call status every 10 seconds via the VAPI REST API. This continues until the call status returns “ended,” at which point the workflow proceeds to outcome evaluation.

Outcome-Based Routing

Customer Busy or No Answer: The system triggers a separate “Follow Up” sub-workflow that schedules a retry attempt, ensuring no lead is lost due to timing.

Call Completed: The recording is downloaded and archived to Google Drive. All structured outputs from the conversation are extracted and synced to Salesforce. The lead record is updated with enriched qualification data.

Call Recording Management

Every completed call recording is automatically archived for compliance, quality assurance, and sales team review.

- Recordings are downloaded directly from VAPI's recording URL
- Files are uploaded to a dedicated Google Drive folder
- File naming convention: {FirstName}_{LastName}_{Phone}_call for easy search and retrieval
- Provides an auditable trail of all AI-prospect interactions

Sub-Workflow Integrations

The pipeline orchestrates two additional sub-workflows to extend its capabilities beyond the core engagement sequence:

Track Emails: Triggered after the initial email is sent, this sub-workflow monitors email engagement metrics including opens, clicks, and replies. This data feeds back into the lead scoring model to prioritize high-engagement prospects for the sales team.

Follow Up (Retry with Leads): Activated when a voice call goes unanswered or the prospect is busy. This workflow manages a retry queue with configurable timing and attempt limits, ensuring persistent but non-intrusive follow-up until contact is established.

Salesforce CRM Integration

The workflow performs two distinct Salesforce operations at different stages of the pipeline:

Initial Lead Creation (Post-Form)

Immediately after form submission and outreach initiation, a lead record is created with the form data and engagement status flags indicating which channels have been activated.

Enrichment Update (Post-Call)

After the AI voice call completes, the existing lead record is updated with all structured outputs extracted from the conversation. This transforms a basic form submission into a richly qualified lead with business intelligence, budget signals, and qualification status.

Technology Stack

Component	Technology
Workflow Engine	n8n (self-hosted)
Voice AI	VAPI (outbound calling + structured extraction)
Email	Gmail API (OAuth 2.0)
Messaging	WhatsApp Business Cloud API (Meta)
CRM	Salesforce
Form Intake	Website Contact Form (Webhook)
File Storage	Google Drive API
Authentication	OAuth 2.0 (Google), Bearer Token (VAPI), Header Auth (Webhook)

Key Features & Highlights

- Instant triple-channel engagement: email, WhatsApp, and AI voice call triggered within seconds of form submission

- AI voice agent conducts natural qualification conversations with full prospect context
- Structured data extraction from voice conversations directly into Salesforce fields
- Intelligent call retry logic for busy/unanswered calls with sub-workflow orchestration
- Automatic call recording archival to Google Drive with structured naming
- Email engagement tracking through dedicated monitoring sub-workflow
- Two-phase Salesforce sync: initial creation on form submit, enrichment after voice call
- Configurable timing delays to prevent overwhelming prospects
- Polling-based call lifecycle management ensuring no call outcomes are missed
- Fully autonomous operation requiring zero human intervention from form to qualified lead

Results & Impact

- Reduced lead response time from hours to under 60 seconds across all three channels
- Automated voice qualification replaced manual SDR calls, freeing the sales team for high-value activities
- Multi-channel approach increased prospect engagement rates compared to single-channel outreach
- Structured voice data extraction eliminated manual call note-taking and CRM data entry
- Retry automation ensured no leads were lost due to missed calls or busy signals
- Call recording archival provided compliance documentation and training material

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